LAB 2 — THINKING SHEET
Linear Models for Classification

This sheet is for you. Preparing for the exam starts here.
I recommend working in pairs, trying to explain your point of view to each other.
Running the code is useful, but it is not the goal of this lab. The goal is to practise predicting, observing, explaining, and experimenting.
For each investigation: PREDICT → RUN & OBSERVE → EXPLAIN → EXPERIMENT
Write short answers. A few words, a sketch, an equation, or a small graph may be enough.

1 — Train/test split

PREDICT — before running

Would taking the first half of the Wine dataset for training and the second half for testing be a good idea? Why?

No, it should be 70/30 in order to efficently train and test the model

RUN & OBSERVE → EXPLAIN

What did the class labels reveal? Why does shuffling matter?

Without shuffling the distribution is much different between train and test set. Shuffling allows to have similar distributions so the training set reflects the real world

EXPERIMENT

Compare shuffled and unshuffled splits. We predicted: _____

We observed / explain: _____

2 — Feature scaling

PREDICT — before running

After fitting StandardScaler on training data, predict the mean/std of scaled training and test features.

Training: 0, 1 Test: ~ 0, 1

RUN & OBSERVE → EXPLAIN

What did you observe? Why must the scaler be fitted only on the training set?

As stated above. Because my model is developed on training data, to correctly interpretate new data I must use the same parameter/rules

EXPERIMENT

Try fitting the scaler on all the data. Does the code work? Why is this methodologically wrong?

It works, It's wrong because we can't normalize our values on future unknown data.

3 — Perceptron: training and test performance

PREDICT — before running

Which error do you expect to be lower? Must this always be true?

The training error, yes it must always be true

RUN & OBSERVE → EXPLAIN

Training error: 0.0 Test error: 0.0449 What does each tell you?

The perceptron algorithm, as a form of ERM, always tries to lower as much as possible the training error fitting the training data. Since the test data is typically different from the training data, our model will obtain an higher prediction error on such unknown data.

4 — Amount of training data

PREDICT — before running

Sketch or describe training/test error as training-set size increases.

The training error doesn't necessarily decrease, but the test error does since the model has more data (so the training set should better represent the real world)

RUN & OBSERVE → EXPLAIN / EXPERIMENT

What differed from your prediction? Change the data permutation: what remains robust?

(*) The training error remains the same, the test error still decreases in an irregular way (but the curve is significantly different)

The training error sits constantly at zero while the test error decreases with an irregular oscillation

5 — Same data, different Perceptrons

PREDICT — before running

With different random states, must coefficients, predictions and errors be identical?

No

RUN & OBSERVE → EXPLAIN

What changed? How can different parameter vectors be plausible classifiers?

Different coefficients, predictions and errors. Both models tends to similar results but with little differences → a different path in the minimization of the training error might have generated those differences. Both models are fitted on the same dataset with the same logic → both are plausible

6 — Does scaling matter for the Perceptron?

PREDICT — before running

What will change if you train the Perceptron on unscaled features?

RUN & OBSERVE → EXPLAIN / EXPERIMENT

Compare scaled/unscaled results; rescale one feature. What happened and why?

7 — Perceptron vs Logistic Regression

PREDICT — before running

Both are linear classifiers. Must they learn the same coefficients, predictions or test error?

RUN & OBSERVE → EXPLAIN

What differs? What extra output does Logistic Regression offer?

Why can two linear classifiers learn different decision rules?

EXPERIMENT

Change training size, split or Logistic Regression C. We changed: _____

We predicted: _____ We observed: _____

END-OF-LAB — CLOSE THE NOTEBOOK

Work individually first; then compare with your partner.

1. Why do we need a test set, and why must preprocessing use training statistics?

2. How can Perceptron and Logistic Regression both be linear yet learn different models?

3. One result I could explain at the exam without running code: